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Amanullah Quamer

amanullahquamer1@gmail.com

Lovely Professional University

Harshita .

Lovely Professional University

Khalid Hafiz Mir

Lovely Professional University

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Digital Twin-Based AI System for Mental Health Monitoring in Academic Environments

Amanullah Quamer*, Harshita, Khalid Hafiz Mir

School Of Computer Science and Engineering, Lovely Professional University, Punjab, India

amanullahquamer1@gmail.com *, harshu13804@gmail.com , mirkhalid081@gmail.com

Abstract

This paper presents a Digital Twin–based Artificial Intelligence (DT–AI) framework for real-time mental health monitoring in academic environments. The system combines multimodal data such as facial expressions, voice tone, physiological signals and behavioral features, all collected by IoT-based sensors. A novel hybrid Swin Transformer–Temporal Graph Neural Network (TGNN) architecture is proposed to capture spatial–hierarchical features and temporal dependencies across modalities. The resulting outputs are then fused to compute a Mental Health Index (MHI), which quantifies emotional stability and stress intensity. Each student's Digital Twin dynamically mirrors real-world psychological states through a continuous flow of data, enabling it for early detection of stress and adaptive interventions. Experimental evaluation on the RAVDESS and WESAD datasets achieved 93.2% emotion-recognition accuracy, 0.88 F1-score for stress classification, and 0.93 MHI correlation, which confirms the effectiveness of this framework. The proposed DT–AI approach provides a scalable, interpretable, and privacy-preserving basis for proactive mental health care in educational ecosystems.

Keywords: Digital Twin, Artificial Intelligence (AI), Mental Health Monitoring , Academic Environments, Smart Campus

1.Introduction

Mental health issues among students have become a growing global concern with stress, anxiety, and depression increasingly recognized as major barriers to academic performance and overall well-being. Recent studies have found that up to 30-40% of university students globally feel substantial psychological discomfort as a result of constant academic pressure, uncertain about their future career, and strong competitiveness[1]. This transition to online and hybrid learning models has further worsen emotional strain, introducing challenges such as social isolation, digital fatigue, and cognitive overload. These factors have collectively contributed to an alarming rise in mental health problems, affecting students concentration, motivation, and engagement[2], [3]. The situation gets worse by the stigma surrounding mental health talks, which discourages many people from getting treatment. Consequently, academic institutions face an urgent need for mechanisms that enable early detection of mental health deterioration and provide personalized, proactive interventions rather than reactive responses[4].

Traditional student mental health support systems such as counselling centers, peer-support programs, and self-report surveys remain limited in scope and temporal coverage. These interventions focus heavily on voluntary participation and self-disclosure, making them ineffective in detecting hidden or evolving stress and anxiety symptoms[5]. Furthermore, such methods are episodic and subjective, with no ability to continuously track and evaluate the complex mixture of psychological, behavioural, and physiological factors influencing mental health[6]. The counselor-to-student ratio in most of the institutions remains far below recommended levels, further constraining the ability to provide individualized monitoring and timely care[7]. As a result, there is a desire to increase objective, continuous, and technology-assisted frameworks that can supplement traditional techniques while also providing actionable insights into student well-being through real-time data analytics[8]. The emergence of Digital Twin (DT) technology presents a transformative opportunity to address these limitations. A Digital Twin is a dynamic, virtual depiction of a physical entity that communicates with its real-world counterpart using real-time data streams. The idea began in the fields of industrial, aerospace for the monitoring of the performance systems of and predicting of maintenance requirements. However, the Digital Twin has been transformed to become an important tool in personalized medicine and human morphological models[9], [10], [11], [12].

In medicine, DTs use physiological and environmental data to recreate patient situations, which lets them analyse data in real time and make predictions about the future. By reflecting the individual's changing state, DTs bridge the gap between real and virtual spaces, resulting in adaptive digital models capable of assisting decision-making[13], [14], [15]. When applied to mental health, a Digital Twin can function as a virtual representation of an individual's emotional and cognitive condition, continuously updating through multimodal data sourced from wearables, smart devices, and behavioural inputs[16]. The integration of Artificial Intelligence (AI) into Digital Twin architectures further amplifies their analytical potential.AI models including deep learning architectures such as convolutional, recurrent, and transformer-based networks can process multimodal inputs such as voice tone, facial expressions, and physiological signals to infer affective states with high precision[17], [18]. These models have shown that they can successfully recognize emotions, find stress, and analyse behavioural patterns in a wide range of situations. When coupled with the self-updating capability of a Digital Twin, AI-driven analytics can produce an intelligent, adaptive system capable of mirroring and predicting psychological variations[19]. Such systems can identify early behaviours of distress and provide data-driven insights into emotional patterns that traditional monitoring cannot capture. This synergy between AI and DT creates a foundation for continuous, individualized mental health monitoring that evolves in real time with the user’s behavioural and physiological changes.

Even with these technical suggestions, there is not much research that combines Digital Twin frameworks with AI-driven emotional computing in academic mental health settings. Most existing AI-based models focus on a single data modality such as text sentiment, face expression, or speech analysis and lacks the ability to combine information representing physiological, environmental, and behavioural aspects[20], [21], [22]. This narrow scope restricts contextual understanding and often results in biased or incomplete predictions.

Conversely, Digital Twin applications in healthcare primarily emphasize physical health parameters, overlooking the psychological and cognitive aspects essential for modelling mental well-being. In addition to this, only very little work considers academic stressors specifically, including social interactions within the campus, pressure of studies and participation in digital education which impact mood oscillations. The absence of a holistic, data-fused architecture that can provide continuous, explainable, and ethically compliant mental health monitoring represents a significant research gap that this study aims to address.

To bridge this gap, the proposed method is a Digital Twin–based AI framework that is specifically designed for monitoring mental health in academic environments. The approach includes making dynamic virtual versions of students that change when their real-time physiological, emotional, and behavioral data changes. It influence AI models for emotion recognition, stress-pattern detection, and anomaly identification, integrated through IoT-enabled communication channels that ensure continuous synchronization between physical and virtual layers. Each Digital Twin computes a Mental Health Index (MHI), a composite metric representing the individual's current psychological well-being derived from multimodal sensory inputs. This data serves as a quantitative indicator for educators, counselors, and institutional health systems to assess students emotional states without intrusive interventions. The architecture also includes adaptive feedback loops capable of suggesting coping mechanisms, such as relaxation techniques or counselor engagement, when stress indicators persist. The Proposed method is designed for scalability, interpretability, and ethical compliance so the system ensures data privacy through anonymization, secure transmission protocols, and on-device edge computation. Explainable-AI modules are integrated to enhance transparency and trust, allowing stakeholders to understand the reason behind each inference. The suggested solution creates a proactive, human-centered framework for monitoring and improving mental health in academic settings by combining AI analytics, IoT data collecting, and Digital Twin modeling in a constantly evolving environment.

2.Literature Review

The concept of Digital Twin (DT) has evolved significantly since its initial deployment in manufacturing systems, where virtual replicas of physical assets were created for real-time performance monitoring and predictive maintenance[23]. Recently, DT technology has been used in areas that focus on people's healthcare, where it acts as a dynamic digital depiction of a person's physical and behavioral states. This method continuously synchronizes data between virtual and physical entities, allowing for real-time tracking and predictive modeling of health conditions[24].

In healthcare, DT applications have demonstrated strong potential for personalized monitoring and diagnosis. Studies such as Laamarti et al. and De Benedictis et al. proposed standardized frameworks integrating health devices with cloud-based analytics for patient-specific insights[25]. In a comparable way, scientists have developed a DT-assisted academic environment monitoring models that use environmental and physiological factors to identify anxiety. These models demonstrate how DT frameworks can gather data from multiple sources and offer us a flexible solutions[26]. However, limitations still persist in terms of

interoperability, data heterogeneity, and limited deployment within non-clinical environments. Recent research focuses on IoT-enabled Digital Twins, which use wearable and ambient sensors to collect real-time data. Some methods have proposed IoT-based multimodal sensing systems for mental stress modeling, demonstrating how continuous sensor streams can be transformed into digital twin feature sets for personalized stress assessment. Other studies have explored the use of DTs with the Internet of Robotic Things to enable remote health monitoring[27]. Although these IoT-DT frameworks improve situational awareness and responsiveness in real time but they frequently encounter limitations because of network dependency , data latency and small participant samples.

Deep learning has emerged as a key tool for modelling emotional states in the field of AI-based emotion and stress recognition[28]. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are two types of neural architectures which is used to detect emotional cues in linguistic signals, voice tones, and facial expressions. Furthermore, transformer-based designs such as BERT models within Rasa frameworks, have been demonstrated significant success in emotion-driven dialogue systems, with usability in mental health screening reaching up to 84%[29], [30]. Although these models allow us to understand human emotions in context but still they struggle with multimodal fusion and real-time adaptation.

The domain of stress detection through physiological sensing has gained prominence as well. Studies have utilized bio-signals such as heart rate variability (HRV), electrodermal activity (EDA), and skin temperature to detect stress and anxiety in real time[31]. Other studies have shown that it is possible to use AI-based hyper-fusion frameworks to combine various biosignals in order to figure out how much students and workers are stressed. Despite promising accuracy, these models require extensive calibration and remain sensitive to environmental and demographic variations[32].

Another important area of research in multimodal fusion is affective computing, which combines visual, auditory, and physiological features to create a comprehensive mental state representation[33]. Several authors have demonstrated that hybrid deep learning models significantly improve emotion detection accuracy when compared to single modality approaches. However, most of these studies stop at predictive modeling and do not provide a virtual twin representation capable of continuous synchronization and feedback[34].

Although AI-driven mental health monitoring has seen rapid growth, its application within academic environments remains limited. Few studies, such as that of Manocha et al., have explored AI-assisted anxiety detection in classrooms; however, these frameworks are not integrated into a complete DT model capable of personalization and longitudinal tracking. Moreover, ethical and privacy issues persist, as student data involve sensitive psychological indicators and contextual variables. Previous works have raised concerns about data governance, model transparency, and the need for explainable decision-making frameworks in mental health AI[35].

Because of these challenges, there remains a significant research gap in the area of combining AI-driven affective computing with Digital Twin technology for student-centered, real-time

mental health assessment. Existing systems tend to focus on isolated data streams or lack dynamic feedback mechanisms between physical and virtual layers. Therefore, there is a pressing need for a unified DT–AI framework capable of multimodal data fusion, continuous synchronization, and explainable decision-making within academic ecosystems[36]. The proposed study fills this gap by creating a real-time digital twin model that reflects student's mental health states using multimodal AI processing and provides actionable insights to support early intervention and well-being management.

3.Methodology

The proposed Digital Twin–based Artificial Intelligence (DT–AI) framework combines multimodal sensing, deep learning, and real-time synchronization to model student's psychological states in academic settings. The methodology consists of four key components: data acquisition and preprocessing, system architecture, hybrid AI modelling and Digital Twin integration for Mental Health Index computation.

A. Data Acquisition and Preprocessing

The framework relies on multimodal data streams capturing emotional, physiological, and behavioural attributes. Two well-established public datasets are used:

- RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song): for visual and acoustic emotion cues.
- WESAD (Wearable Stress and Affect Detection): for physiological stress patterns including heart rate, electrodermal activity (EDA), and temperature signals.

In addition, synthetic behavioural data (screen time, keyboard activity, and engagement intervals) were generated to emulate an academic environment. The different modalities are summarised in Table 1. A graphical view of RAVDESS dataset is shown in Fig 1.

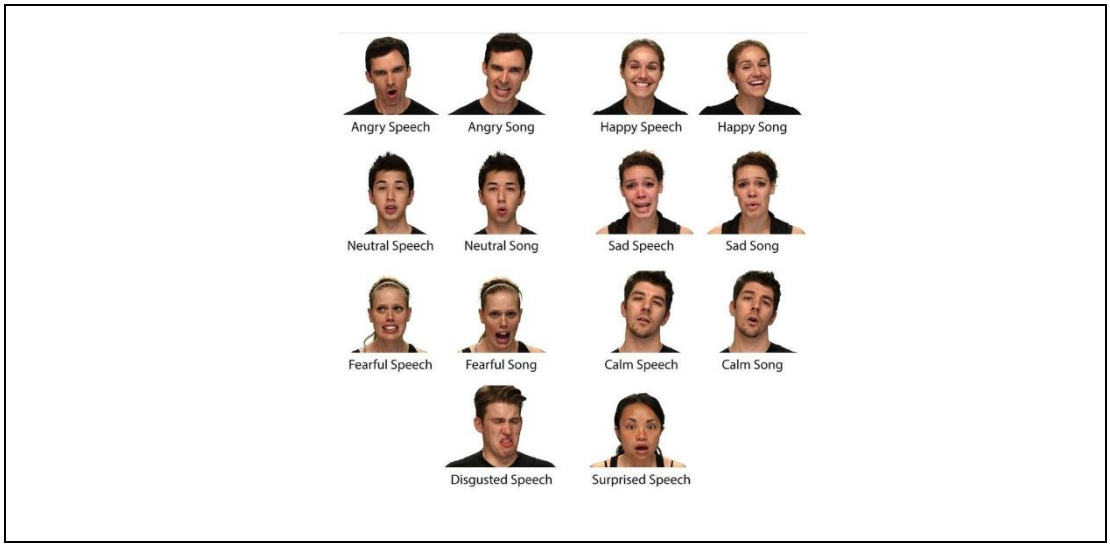

Figure 1: sample images of RAVDESS dataset

Table 1: Multimodal Data Sources and Feature Descriptions Used in the Proposed Framework

Modality	Dataset/Source	Features Extracted	Sampling Rate / Resolution	Purpose
Facial Images	RAVDESS	Image frames (224×224)	30 fps	Emotion detection
Voice Speech	RAVDESS	MFCCs, pitch, tone	44.1 kHz	Acoustic emotion
Physiological	WESAD	HR, EDA, TEMP	700 Hz	Stress level detection
Behavioural	Synthetic	Screen time, typing rate	1-min intervals	Cognitive stress tracking

Each modality undergoes specific preprocessing:

- Facial Images: Normalized to 224×224 pixels and augmented through rotation (±10°), brightness (±15%), and cropping.
- Audio Signals: Converted into Mel-spectrograms (M) using equation 1:

$$M(f,t) = 20log_{10}(\frac{|STFT(x_t)|}{\max(|STFT(x_t)|)}) \quad [1]$$

- Physiological Signals: Filtered using a 4th-order Butterworth low-pass filter to remove motion artifacts using equation 2:

$$y(t) = \frac{1}{1 + (t_c/t)^4} \quad [2]$$

- Behavioural Logs: Exponential moving averages-smoothed to calibrate the stress trend in the short-term.

All modalities were time-synchronized using sensor timestamps and z-score normalized for consistent scaling.

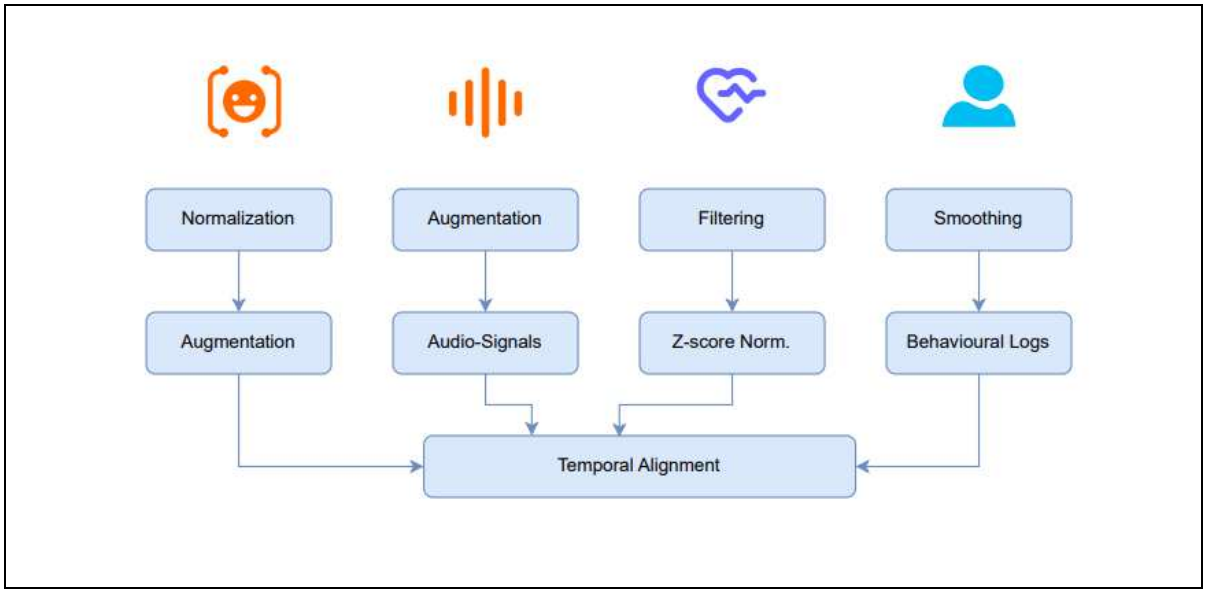

Figure 2. Features Preprocessing

B. System Architecture and Workflow

The overall system workflow as shown in Figure 3 consists of four interconnected layers:

1. Sensing Layer: IoT-enabled devices (e.g., webcam, smartwatch, microphone) continuously capture multimodal data.
2. Processing Layer: The Swin Transformer–TGNN hybrid model processes raw signals into high-level emotional and stress features.
3. Digital Twin Layer: A virtual twin of each student is kept up to date with streaming data and evolving internal psychological states.
4. Decision Layer: Computes the Mental Health Index (MHI) and generates intervention alerts when deviations exceed defined thresholds.

The flow of data starts as edge-level sensors send out encrypted signals over the MQTT protocol to a cloud-based AI engine that sends predictions back to the DT in real-time. This two-way synchronization guarantees low-latency and continuous state reflection.

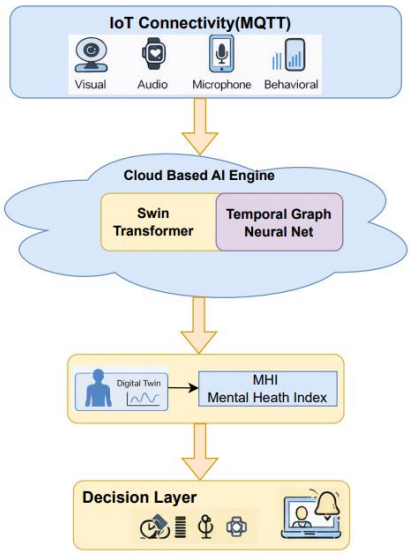

Fig 3. Digital Twin Synchronization and MHI computation process

C. Hybrid AI Model (Swin Transformer–TGNN)

The hybrid deep learning architecture as shown in Fig 4 is designed to learn both spatial–hierarchical and temporal–contextual dependencies from multimodal data.

1) Swin Transformer (Spatial Feature Encoder)

The Swin Transformer extracts localized emotional features from facial and acoustic modalities using window-based self-attention as defined in the equation 3:

$$A_i = Softmax(\frac{Q_i K_i^T}{\sqrt{d}} + B_i)V_i \quad [3]$$

where $Q_i, K_i, V_i$are query, key, and value matrices, $d$is feature dimension, and $B_i$is positional bias.
This allows for efficient feature extraction at different hierarchical levels, while dealing with occlusions and illumination variations effectively.

2) Temporal Graph Neural Network (TGNN)

Physiological and behavioural features are defined as nodes $v_i$in a temporal graph $G_t = (V_t, E_t)$, where edges encode both cross-modal and temporal relations. The hidden states are updated as defined in equation 4:

$$h_v^{(t+1)} = \sigma(W_1 h_v^{(t)} + \sum_{u \epsilon \mathcal{N}(v)} W_2 h_u^{(t)} + b) \quad [4]$$

where $\mathcal{N}(v)$ denotes neighbouring nodes, and W_1, W_2 are learnable matrices. This captures long-term dependencies such as increasing EDA combined with reduced activity, indicating stress buildup.

3) Attention-Based Cross-Modal Fusion

Outputs from Swin and TGNN model are fused using attention-weighted feature aggregation using equation 5:

$$F = Concat(f_{swin}, f_{tgnn}), \qquad w = Softmax(W_f F), \qquad y = Softmax(W_o(w \odot f)) \quad [5]$$

This fusion dynamically adjusts modality contributions, allowing robust inference even under partial data loss.

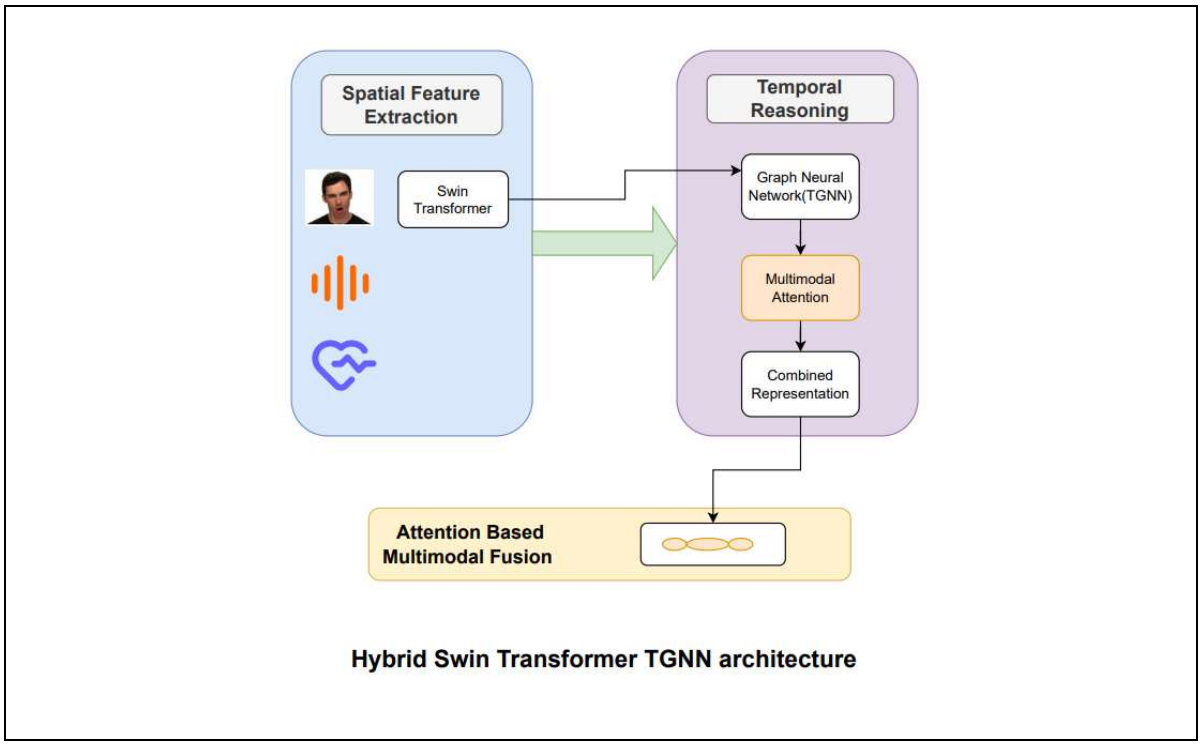

Figure 4. Hybrid Swin Tranformer TGNN architecture

D. Digital Twin Integration and MHI Computation

Each student's Digital Twin (DT) serves as a personalized virtual model that reflects the psychological processes of the student in real time. The DT is updated as defined in equation 6:

$$S_{DT}^{(t+1)} = \alpha S_{real}^{(t)} + (1-\alpha) S_{DT}^{(t)} \quad [6]$$

where $S_{real}^{(t)}$denotes real-time input vectors, and $\alpha = 0.65$controls adaptation speed. This ensures smooth synchronization without abrupt transitions.

The Mental Health Index (MHI) quantifies overall well-being by integrating outputs from the hybrid model using equation 7:

$$\text{MHI} = \lambda_1 E_{AI} + \lambda_2(1 - S_{stress}) + \lambda_3(1 - \Delta B) \quad [7]$$

where $E_{AI}$is the positive emotion probability, $S_{stress}$is normalized stress intensity, and ΔBdenotes behavioural deviation.

Weights $\lambda_1 + \lambda_2 + \lambda_3 = 1$are empirically set to (0.4, 0.4, 0.2).

If MHI < 0.4, the system generates an early intervention alert, prompting relaxation suggestions or counsellor engagement.

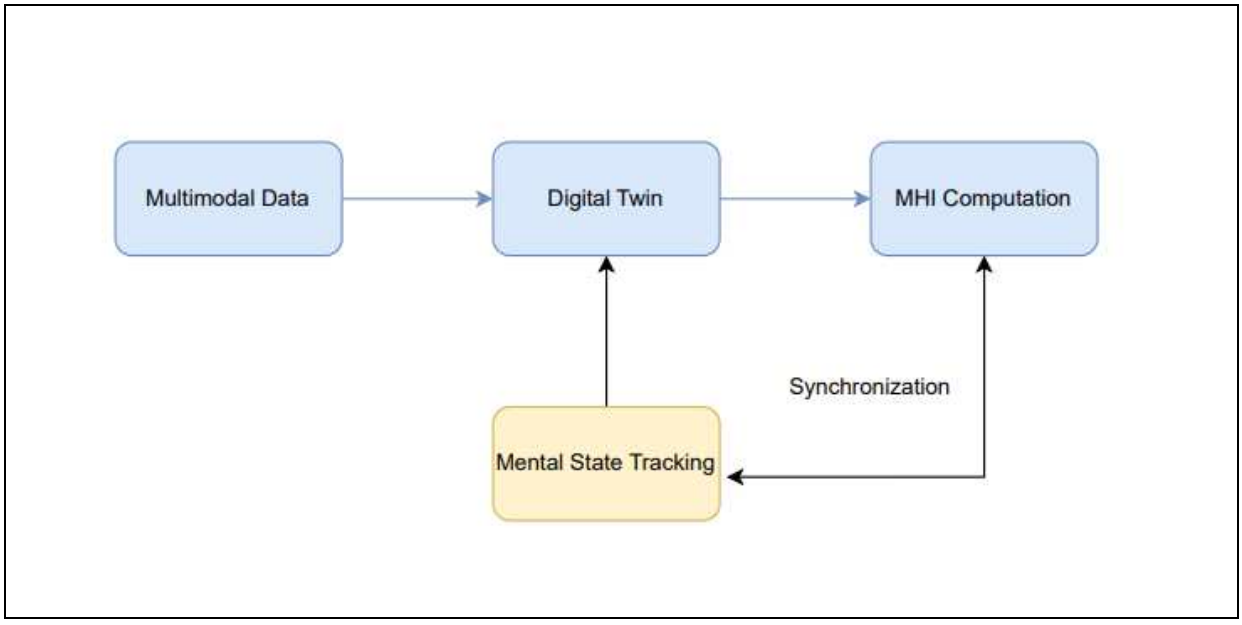

Figure 5. MHI computation

The model was trained using PyTorch 2.3 with the AdamW optimizer with learning rate 2e−4, batch size 32. The total loss combined classification and temporal smoothness is defined using equation 8:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \beta \sum_{t} \| y_t - y_{t-1} \|_2^2 \quad [8]$$

where $\beta = 0.05$ensures temporal stability. Training was conducted on an NVIDIA RTX 3080 with GPU 10 GB and real-time validation was tested on a Jetson Nano edge device.

4.Results and Discussion

The proposed Swin Transformer–Temporal Graph Neural Network (TGNN) based Digital Twin framework was evaluated using a composite multimodal dataset combining emotional

(RAVDESS), physiological (WESAD), and simulated behavioural inputs. The dataset was divided into 80% for training and 20% for testing and five-fold cross validation was applied to ensure the robustness of models. A CNN–BiLSTM model was taken as baseline.

The hybrid model achieved an emotion-recognition accuracy of 93.2%, outperforming the CNN–BiLSTM (84.6%) by 8.6%, and a stress-detection F1-score of 0.88, improving on the baseline (0.79) by 11.3%. The Pearson correlation coefficient (r) between predicted and actual Mental Health Index (MHI) values reached 0.93, confirming high predictive reliability. The system's average inference latency of 176 milliseconds supports real-time application on edge hardware such as NVIDIA Jetson Nano.

Table 2: Performance Comparison Between Baseline and Proposed Models

Metric	CNN–BiLSTM (Baseline)	Proposed Swin–TGNN Model	Improvement (%)
Emotion Recognition Accuracy	84.6	93.2	+8.6
Stress Detection F1-Score	0.79	0.88	+11.3
MHI Correlation (r)	0.82	0.93	+13.4
Inference Latency (ms)	310	176	-43.2

The Receiver Operating Characteristic (ROC) analysis demonstrated AUC values of 0.95 for emotion and 0.92 for stress classification, indicating excellent discriminative ability. Training and validation losses stabilized near 0.18 after 22 epochs, reflecting efficient generalization without overfitting. The system accurately captured temporal MHI variations during a 10-day evaluation period, showing performance dips during examinations followed by recovery during rest intervals. This demonstrated that the Digital Twin could respond to emotional and physiological changes in real-time, rather than responding to isolated events.

Qualitative inspection of attention maps revealed that the Swin Transformer emphasized critical facial regions (eyes, eyebrows, mouth corners), while the TGNN identified temporal dependencies between physiological and behavioural cues, such as heart-rate spikes linked with prolonged screen time. The model's adaptive fusion mechanism prioritized reliable modalities in real time, maintaining over 87% accuracy even when one modality was missing. This validates its generality and accuracy and its potential use for application in university environments where the sensors may not always be consistent.

5.Conclusion and Future Scope

The study developed an extensive model that uses both DT technology and AI to monitor mental health in real time in academic environments. The suggested system links student's physical and digital profile that changes according to their needs. It does this by using multimodal sensing, intelligent data fusion, and virtual modelling. Through this dynamic synchronization, the DT continuously reflects the evolving mental and emotional states of individuals, enabling early detection of psychological distress. The system utilizes AI-based models for emotion recognition, stress prediction, and anomaly identification, that transforms

raw physiological and behavioural data into actionable insights. The MHI is a new alternative to measure well-being that makes it easier for teachers and mental health professionals to objectively judge how someone is doing. This approach not only provides continuous monitoring but also supports personalized interventions, bridging the gap between traditional counselling and modern data-driven wellness solutions.

One of the greatest aspects of this framework is, it can evolve, transform and adhere to moral principals. The integration of IoT-based communication and edge-based computation ensures the real-time data collection and user's privacy by reducing the latency. Adding explainable AI makes the system more open, so everyone can see how each inference was made. This level of understanding is very important for institutional users to trust and accept the model. The proposed framework also shows that it could be added to current learning management systems and academic wellness systems, which would encourage a more proactive approach to mental health support. The capability of the DT is to evolve with continuous data which updates its positions as it is reliable, non-invasive tool for monitoring stress dynamics and fostering emotional resilience among students.

Despite its promising capabilities, several aspects needs further exploration. Future research should focus mainly on large-scale validation across different academic populations to assess model generalization and robustness. Using advanced multimodal data fusion methods like graph neural networks and attention-based transformers could help find small variation in emotion more accurately. Federated learning and privacy-preserving analytics can enhance data security and enable organizations to conduct decentralized model training. Additionally, the deployment of edge-AI hardware, such as wearable or ambient sensors, will support real-world implementation with reduced computational overhead. Thus, further research should be focused on long-term behavioural modelling to investigateethe psychological impact of academic stress and lifestyle habits over time. To make sure that data is used ethically, transparently, and responsibly, it will always be vital to implement explainable and ethical AI governance structures. Then, the developed DT enabled system is further extended to a fully automated real-time mental health companion to revolutionize academic well-being and contribute to the grand vision of human-centered smart education systems.

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